

IMSC25: Original Triangle Problems

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Unless stated otherwise, triangles ABC are considered to be acute and scalene with $AB < AC$.

EMO stands for Estonian Mathematical Olympiad, OR for Open Round (autumn and winter contests, respectively), ETST for Estonian Team Selection Test and BW for Baltic Way international competition. Star (*) indicates that the statement has been changed.

✓ 1a. (EMO24-9) In triangle ABC , the perpendicular bisector of BC intersects lines AC and AB at points D and E , respectively. The circle with diameter DE intersects lines AC and AB again at points K, L , respectively ($K \neq D, L \neq E$). Let M be the midpoint of BC . Prove that points K, L, M are collinear.

2a. (EMO25-8)* In the right triangle ABC , M is the midpoint of the hypotenuse AB . Point D is chosen on the leg BC so that the line segment DM meets (ACD) again at K ($K \neq D$). Let L be the reflection of K in M . The circles (ACD) and (BCL) meet again at N ($N \neq C$). Prove that $ACKN$ and $BCNL$ are trapezoids.

✓ 1b. (EORa23-S) In triangle ABC , altitudes BE and CF intersect at H . The tangent to (ABC) at A intersects (AEF) again at $T \neq A$. $(TBE), (TCF)$ intersect again at $K \neq T$. Prove that $\angle KHB = \angle ABC$.

2b. (EORa24-S)* In triangle ABC , let O be the circumcenter. Points D, E are chosen on side BC such that AD is the altitude in $\triangle ABC$ and AE is the angle bisector in $\triangle ACD$. The internal angle bisectors of $\triangle AOB$ meet at J . Let K be the projection of J onto AC . Prove that $BE = 2JK$.

3b. (EORw24-S) In triangle ABC , the extension of the altitude AD of $\triangle ABC$ beyond D intersects (ABC) at E . Let F is the midpoint of side CE . The circles (ABC) and (DEF) intersect again at $G \neq E$. Let P be the projection of A onto GF . Prove that $DA = DP$.

4b. (EMO25-12)* In triangle ABC , the altitudes AD, BE and CF intersect at H . Let M be the midpoint of BC . Point K is chosen on the extension of EM beyond M and point L is chosen on the segment FM such that $MK = ML = MD$. Let P be the projection of D onto HM . Prove that PD bisects $\angle KPL$.

5b. (ETST25) In triangle ABC , I is the incenter. Line AI intersects (ABC) at D ($D \neq A$) and side BC at E . Lines BI, CI intersect $(CDI), (BDI)$ again at points K, L , respectively ($K \neq I, L \neq I$). Prove that points A, K, L, E are concyclic.

6b. (ETST25) In triangle ABC , $\angle ABC > 90^\circ$ and $AB < BC$. The incircle of $\triangle ABC$ with center I touches the sides at points D, E, F , respectively. Line AI intersects side BC at K . Rays IB, IC intersect $(KIF), (KIE)$ again at points P, Q , respectively ($P \neq I, Q \neq I$). Prove that line PQ bisects DK .

1c. (BW24-P12) In triangle ABC with circumcircle ω , let M be the midpoint of the arc BC of ω containing the point A , and let $X \neq M$ be the other point on ω such that $AX = AM$. Points E and F are chosen on sides AC and AB of the triangle ABC such that $EX = EC$ and $FX = FB$. Prove that $AE = AF$.

2c. (EMO24-12) In triangle ABC , the internal angle bisectors BE and CF intersect at I . Let M, N be the midpoints of BI, CI , respectively. Line FM intersects the external angle bisector of $\triangle ABC$ at vertex B at K . Similarly, line EN intersects the external angle bisector of $\triangle ABC$ at vertex C at L . Prove that points B, C, K, L are concyclic.

3c. (BW24-P14) In triangle ABC with circumcircle ω , the altitudes AD, BE and CF intersect at point H . A point K is chosen on the line EF such that $KH \parallel BC$. Prove that the reflection of H in KD lies on ω .

1d. (AOPS) In triangle ABC , let D be the foot of the altitude from A . Let J be the reflection of the incenter I of $\triangle ABC$ in line AD . Lines BI, CI intersect $(BDJ), (CDJ)$ again at points $K \neq B, L \neq C$, respectively. Let H be the orthocenter of $\triangle KJL$. Show that $\angle AIH = 90^\circ$ and $DH = DI$.

2d. (AOPS) In triangle ABC , let AD, BE and CF be the altitudes. Points K, L are chosen outside $\triangle ABC$ on rays BE, CF , respectively, such that $\angle AKC = 90^\circ$ and $\angle ALB = 90^\circ$. (DKL) intersects BC again at $X \neq D$. Prove that BC is tangent to (XEF) .

1 Bonus: tangent to each other (TTEO)

I filled this second page with some problems about tangent circles, in which the point of tangency is not given.

- 1t. (Folklor) Prove that in triangle ABC , the incircle and the nine-point circle are TTEO (Feuerbach point).
- 2t. (SL2020) Let $ABCD$ be a convex quadrilateral with $\angle ABC > 90^\circ$, $\angle CDA > 90^\circ$ and $\angle DAB = \angle BCD$. Denote by E and F the reflections of A in lines BC and CD , respectively. Suppose that the segments AE and AF meet the line BD at K and L , respectively. Prove that (BEK) and (DFL) are TTEO.
- 3t. (SL2020) Let ABC be a triangle with $AB < AC$, incenter I , and A excenter I_A . The incircle meets BC at D . Define $E = AD \cap BI_A$, $F = AD \cap CI_A$. Prove that (AID) and (I_AEF) are TTEO.
- 4t. (IMO 2015 P3) Let ABC be an acute triangle with $AB > AC$. Let Γ be its circumcircle, H its orthocenter, and F the foot of the altitude from A . Let M be the midpoint of BC . Let Q be the point on Γ such that $\angle HQA = 90^\circ$ and let K be the point on Γ such that $\angle HKQ = 90^\circ$. Assume that the points A, B, C, K and Q are all different and lie on Γ in this order. Prove that (KQH) and (FKM) are TTEO.
- 5t. (AOPS, CAMO 3) Let ABC be a triangle with incircle ω , and let ω touch sides BC, CA, AB at D, E, F , respectively. Let M be the midpoint of EF and let DT be the diameter of ω . Line MT meets the line through A parallel to BC at P and ω again at $Q \neq T$. Let $X = DF \cap AP$, $Y = DE \cap AP$. Prove that (APQ) and (DXY) are TTEO.

1.1 Source: Neurogeometry channel

- 6t. In triangle ABC , H_A is the A -Humpty point and E is the projection of H_A onto AC . Let D_E be the E -Dumpty point of $\triangle AEB$. Line AD_E intersects side BC at F . Prove that (AH_C) and (BD_EF) are TTEO.
- 7t. In triangle ABC , the incircle with center I touches side BC at D . Tangent from D to (AI) intersects AI at E . Prove that (ABC) and (DEI) are TTEO.
- 8t. Harmonic quadrilateral $ABSC$ is given. Let M be the midpoint of BC and let D be a point on AS . Define $P = DM \cap AB$, $Q = DM \cap AC$. Prove that (APQ) and (SMD) are TTEO.
- 9t. In triangle ABC , the incircle with center I touches side BC at D . Let K, L be the A -South and A -North poles of $\triangle ABC$, respectively. Let M be the midpoint of IK and let N be the midpoint of LM . Line KD intersects (ABC) again at S ($S \neq D$). Prove that (ISD) and (AMN) are TTEO.
- 10t. In triangle ABC , let I_A, I_B, I_C be the excenters. Let P, Q be the orthocenters of $\triangle BAI_C, \triangle CAI_B$, respectively, and define $R = PI_C \cap QI_B$. Prove that $(I_A I_B I_C)$ and (PQR) are TTEO.
- 11t. In triangle ABC , the incircle with center I touches side BC at D . Let M be the A -North pole of $\triangle ABC$. Line MI intersects (BIC) again at K ($K \neq I$). Line ID intersects (ADC) again at L ($L \neq D$). Define $N = BM \cap CL$. Prove that (ACK) and (MNL) are TTEO.
- 12t. In triangle ABC , I is the incenter and H is the projection of I onto the external angle bisector of $\angle B$. Line HI intersects (AIC) again at D ($D \neq I$). Prove that (AIB) and (DCH) are TTEO.
- 13t. In triangle ABC , the incircle with center I touches side BC at D . Line AD intersects the external angle bisectors of $\angle B$ and $\angle C$ at points E and F , respectively. Let F' be the reflection of F in C . Define $G = AD \cap IF'$. Prove that (FEF') and (IDG) are TTEO.
- 14t. In triangle ABC , the incircle with center I touches side BC at D . Line AD intersects the external angle bisectors of $\angle B$ and $\angle C$ at points E and F , respectively. Let M be the midpoint of CM and define $K = FM \cup AI$. Prove that (ACK) and (BDE) are TTEO.